Lesson 12: Hearing It Happen

GameMaker

VIDEO GAME CREATION

This week: your game finally sounds like a game. Four moments - firing, hitting, clearing a wave, and losing - each get their own distinct cue.

Open your own [YourName]Game project from Lesson 11 - don't start a new one.

If you already built an optional sound idea in an earlier lesson (Lessons 2, 3, 4, or 6), keep it - today reuses it directly instead of replacing it. If you didn't, today's steps are written as if starting fresh.

Setup

1. Open your [YourName]Game project from Lesson 11.
2. Create one new Sound resource: sfx_wave - today's one genuinely new sound. Keep sfx_win, sfx_lifelost, and sfx_gameover from Lessons 6 and 8 if you already have them.

Level 1 - Minimal (MVP)

Four sounds fire at the correct moment, at least once each: a shot, a hit/explosion, a wave-clear (on any non-final wave), and a game over. No silence, no obviously wrong triggers.

3. Confirm sfx_shoot already plays via audio_play_sound() in your shoot input code (Lesson 3) - add it now if it's missing.
4. Confirm sfx_hit already plays via audio_play_sound() in obj_bullet's Collision event (Lesson 4/5) - add it now if it's missing, right after the score/enemies_remaining updates.
5. In obj_bullet's Collision event's nested if (from Lesson 11), add audio_play_sound(sfx_wave, 1, false) to the wave-advance (else) branch.
6. On obj_enemy's Step event, confirm audio_play_sound(sfx_gameover, 1, false) already fires inside the lives-reaches-zero branch (from Lesson 8) - add it now if it's missing. This sound is mandatory starting today even if you skipped it before.
7. Press F5 (Run) and play through a full round - shoot, hit, an early wave clear, and a loss should each sound distinct.

Highlighted lines are new this week; everything else is code you already have.

```
// obj_bullet's Collision event with obj_enemy (revised):
audio_play_sound(sfx_hit, 1, false);
obj_gamemanager.enemies_remaining -= 1;
obj_gamemanager.__dnd_score += 10;
if (obj_gamemanager.enemies_remaining <= 0) {
    if (obj_gamemanager.wave >= obj_gamemanager.total_waves) {
        obj_gamemanager.has_won = true;
```

```
        obj_gamemanager.game_active = false;
        audio_play_sound(sfx_win, 1, false); // your Lesson 6 Level 3
sound
    } else {
        obj_gamemanager.wave += 1;
        obj_gamemanager.enemies_remaining = 5;
        audio_play_sound(sfx_wave, 1, false);
        spawn_enemies();
    }
}
instance_destroy(other);
instance_destroy(self);

// obj_enemy's Step event (revised):
if (!obj_gamemanager.game_active) exit;
y += enemy_speed;
if (y > room_height) {
    obj_gamemanager.__dnd_lives -= 1;
    if (obj_gamemanager.__dnd_lives <= 0) {
        obj_gamemanager.has_lost = true;
        obj_gamemanager.game_active = false;
        audio_play_sound(sfx_gameover, 1, false); // your Lesson 8 Level 3
sound, mandatory as of today
    } else {
        audio_play_sound(sfx_lifelost, 1, false); // your Lesson 8 Level 3
sound
    }
    instance_destroy();
}
```

Alternate Path: GML Visual (no typing)

The same sound calls work as drag-and-drop blocks instead of typed code.

8. On obj_bullet's Collision event (GML Visual), add a "Play Audio" block (sfx_hit) right after your existing score/enemies_remaining Variable blocks.
9. Add two more "Play Audio" blocks to the win-check's If Variable block from Lesson 11 - sfx_win on the True output, sfx_wave on the False output.
10. On obj_enemy's escape branch, confirm a "Play Audio" block (sfx_gameover) already sits inside the lives-reaches-zero If Variable block, with sfx_lifelost on the Else (from Lesson 8) - add them now if missing.

Level 2 - Complete (Alpha/Beta)

All four sounds fire reliably every time, with no missing or duplicate playback, and each is clearly distinguishable from the others by ear.

- Play a full round: shoot, hit, an early wave clear, and a loss each produce a distinct sound.
- If you built the optional win sound, confirm it still plays on the true win, and sfx_wave doesn't also fire on that same frame.
- audio_play_sound() calls layer naturally - confirm rapid shooting and a hit landing together both play correctly, with no sound cutting another off.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Audio: if you haven't already, add the optional win sound now - it's the same audio_play_sound(sfx_win, ...) pattern Lesson 6 built, alongside today's sfx_wave.
- Audio: add slight pitch variation to hits with audio_play_sound(sfx_hit, 1, false) followed by audio_sound_pitch(inst, random_range(0.9, 1.1)), where inst is the sound's returned instance handle.
- Structure: group all your Sound resources under one folder in the Asset Browser for organization, now that there are five of them.

If Something's Wrong

- No sound on wave-clear: confirm the sfx_wave call is actually inside the else branch, not after the if/else.
- Game-over or life-lost sound plays repeatedly: confirm the check is still guarded by "if (obj_gamemanager.__dnd_lives <= 0)" / "else" as a one-time check (from Lesson 8), not re-triggering every Step.
- Wrong instance's variables referenced: confirm the obj_gamemanager. prefix is present throughout the revised blocks, since this code still runs from obj_bullet and obj_enemy.

Want More?

Student Resource List > GameMaker section, or manual.gamemaker.io